

Qatar-1b

R-2230

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120530 · created 2026-08-15T01:06:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456077.90027 +/- 0.00021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.057
Transit depth (Rp/Rs)^2	1.2 +/- 1.24 [%]
Orbital Inclination (inc)	79.55 +/- 2.9
Airmass coefficient 1 (a1)	0.74 +/- 0.44
Airmass coefficient 2 (a2)	0.03 +/- 1.42
Scatter in the residuals of the lightcurve fit is	56.0 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.022 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.11 $\pm$ 0.057 vs 0.1463 (deviation 0.48 σ)
Duration fit vs published	0.022 d vs 0.0692 d (deviation 1.38 σ)
Mid-time O-C	-0.2 min
Depth SNR	1.5
Residual scatter	56.0 %

Light curve

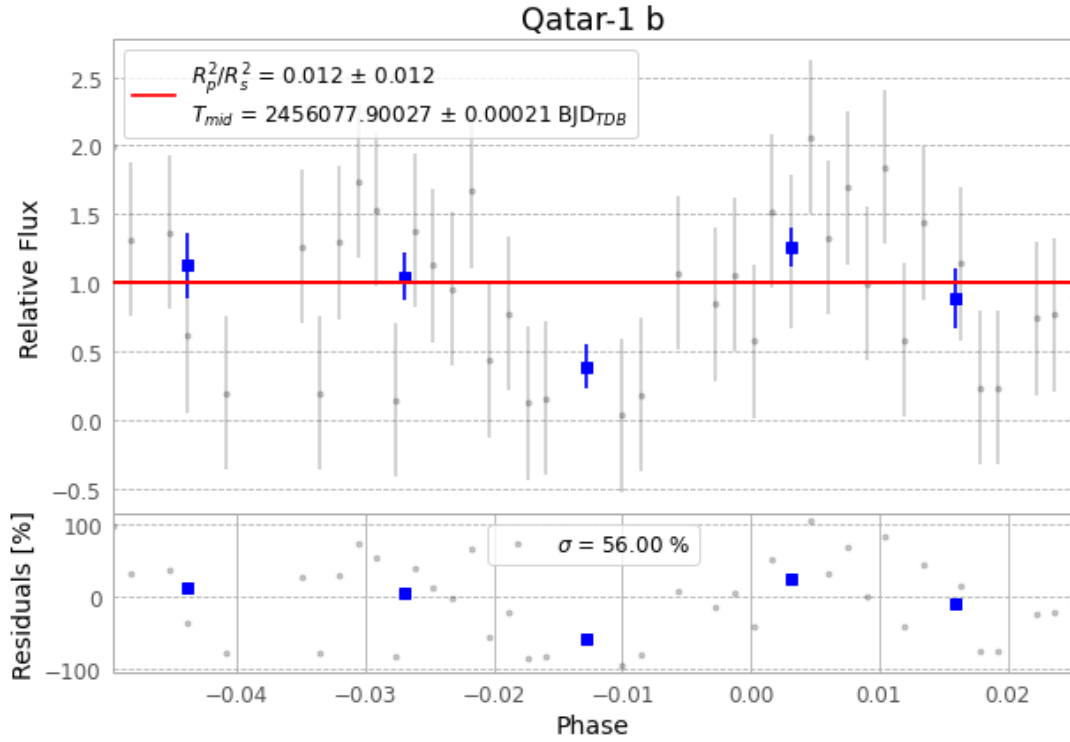


Figure 1 — FinalLightCurve_Qatar-1 b_2012-05-30.png (SHA-256 ae0edcd9645c72c1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 450], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c51a21c90e2eae98...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), Qatar-1120530075413.FITS → Qatar-1120530111813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120530.log (bb45b4ea6691...), FinalLightCurve_Qatar-1 b_2012-05-30.png (ae0edcd9645c...), FinalLightCurve_Qatar-1 b_2012-05-30.csv (11dfa0113908...)

Compute resources

Wall time 138.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:06Z.